TU Wien Informatics

Indoor Air Quality in Sport Facilities

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Context

- Indoor air quality gets far less attention than outdoor pollution, but it matters just as much, especially in sport facilities
- Climbing halls, boulder gyms, and fitness centres combine high occupancy, intense physical activity, and often limited ventilation.
- These conditions can push CO2, particulate matter, and temperature to uncomfortable or even harmful levels.

Relevance

- Poor air quality can affect comfort, focus, and health during exercise
- Facility managers have little data on when to increase ventilation
- This project turns raw sensor data into evidence they can act on

Problem Statement

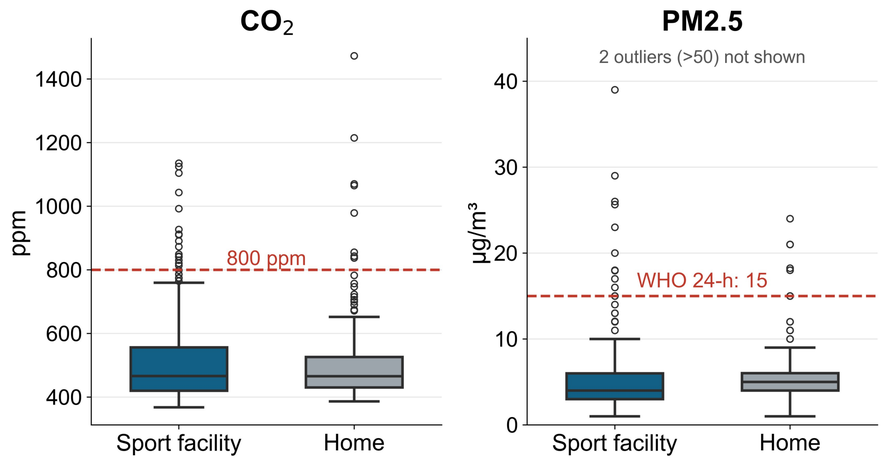
1. How do CO2, PM2.5, temperature and humidity differ between sport facilities and home?
2. Does activity intensity have a measurable effect on CO2 levels?
3. What temporal patterns appear across the day and the study period?

Method

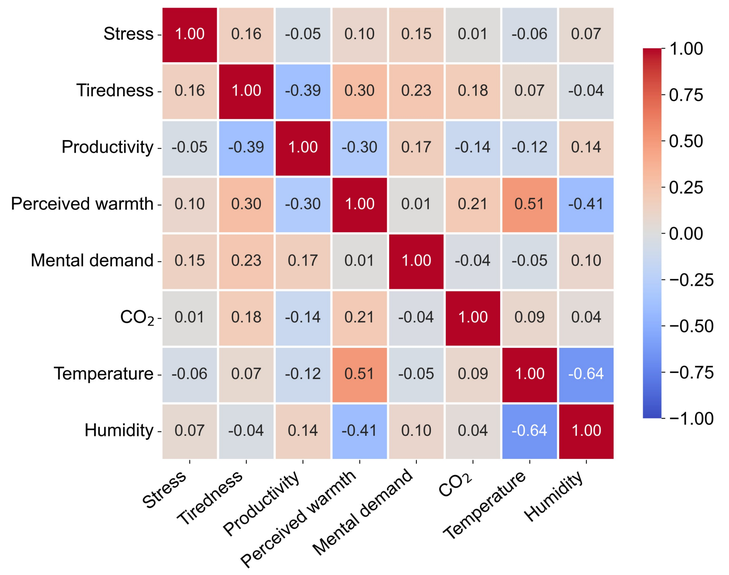
- 17-day study (29 June – 15 July 2025): 22 survey respondents, 20 with sensor data at home and/or at their sport-facility workplace
- Fixed AirControl sensors (CO2, temperature, humidity) and portable AtmoTube sensors (VOC, PM1 / PM2.5 / PM10)
- Daily surveys (scale 1–7): stress, tiredness, productivity, thermal comfort, mental demand
- Sensor data matched to survey times by linear interpolation; Spearman rank correlation for ordinal scores (Pearson as a check)

Main Outcomes

- CO2 is higher at the sport facility at survey times (≈512 vs 500 ppm); readings above 800 ppm are slightly more frequent there (4.6% vs 3.1%)
- PM2.5 is higher at home (5.8 vs 5.1 µg/m³); both are well below the WHO 24-h limit of 15 µg/m³
- Air quality and well-being correlations are mostly weak (up to ±0.18); only perceived warmth clearly follows temperature (ρ = 0.51)
- A questionnaire proxy for activity (tiredness minus productivity) is only weakly linked to CO2 (ρ = 0.20, about 9 ppm per point, 3% of the variance)



CO2 and PM2.5 at sport facilities and at home. CO2 is higher at work, PM2.5 is higher at home.



Spearman correlation matrix: most air quality and well-being correlations are weak.

Conclusion

- Sport facilities raise CO2 through occupancy and exertion, while homes contribute more fine particles
- Short-term air quality fluctuations at these moderate levels show only a weak link to how people feel

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